WESTERN MICHIGAN UNIVERSITY

College of Engineering and Applied Sciences

Course Name: Lab. Intro. To Microprocessors

Course Number: ECE 2510

Instructors:
Christian Blanchard
Pravana Avuthu

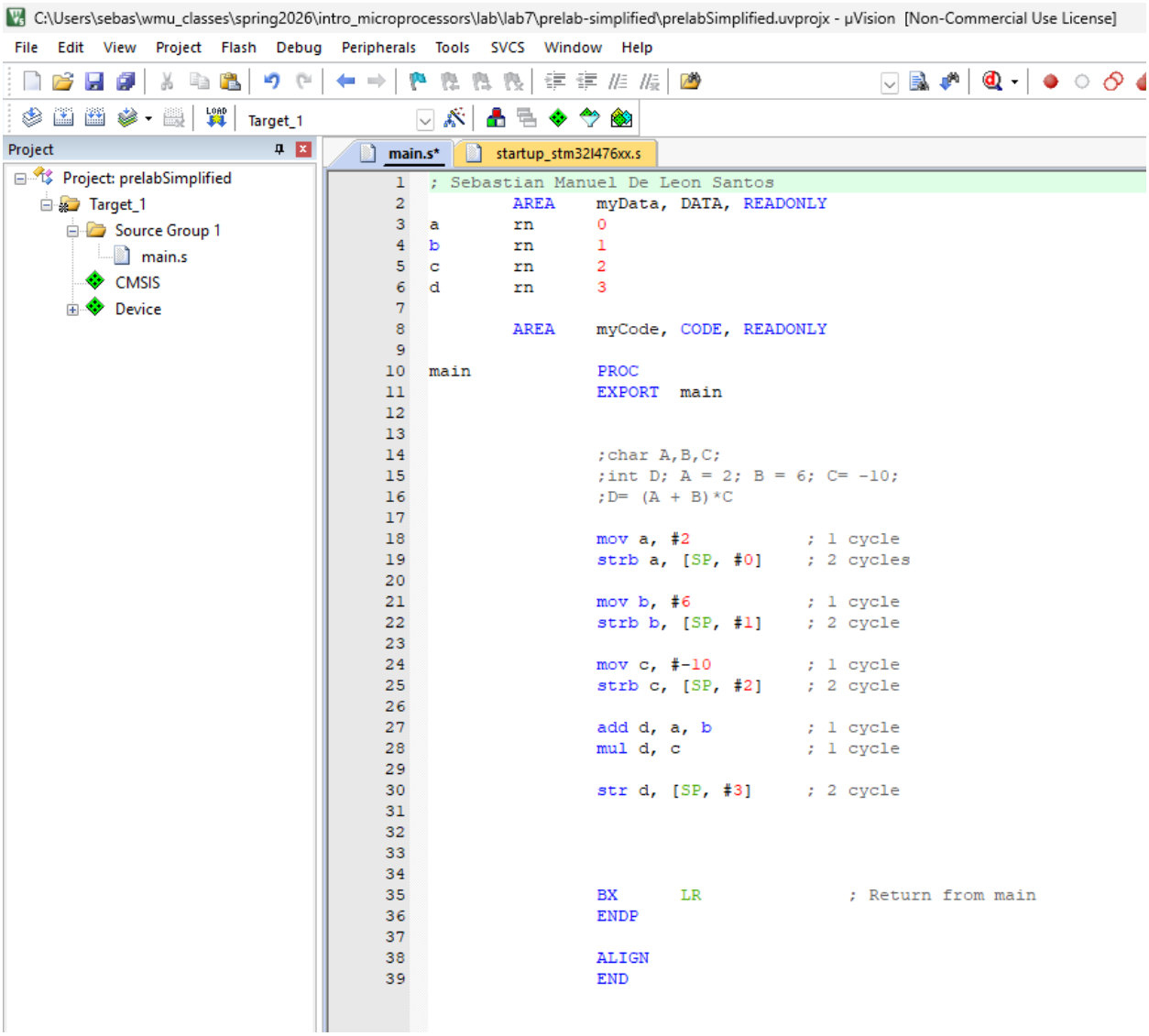
Lab #7 prelab

Student:

De Leon Santos, Sebastian Manuel – WIN: 577866168

Task 1

Calculate your program's total E-cycle count.



Total of 13 cycles

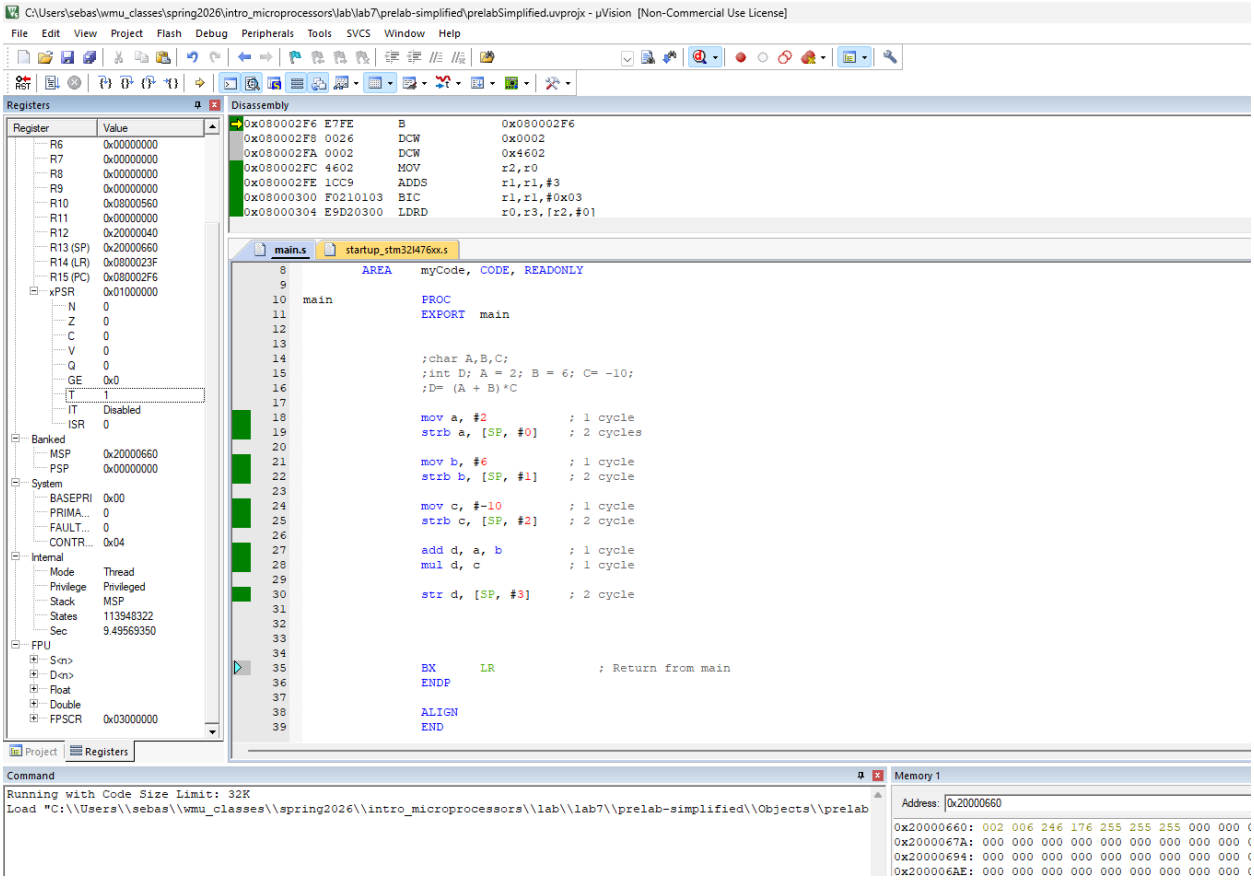


Figure 1: Task 1 code with memory output showing the data stored

Task 2

Given the following program and the instruction reference manual linked in Elearning, calculate the
number of cycles it takes to execute and the execution time.

```
AREA myData, DATA, READWRITE
i SPACE 4

AREA myCode, CODE, READONLY
EXPORT __main
ENTRY

__main
        MOV R0, #37
        LDR R1, =i
        STR R0, [R1]
Loop
        LDR R0, [R1]
        CMP R0, #0
        BEQ Done
        SUB R0, R0, #1
        STR R0, [R1]
        B Loop
Done
        B Done

END
```

counting cycles starting at _main

(1 + 2 + 2 + [(loop label) 2 + 1 + (BEQ not taken) 1 + 1 + 2 + (3)] * 36 times + (CMP) 1 + (BEQ taken) 1 + 1) cycles = 368 cycles

Time in Seconds = 368 cycles * 250 nanoseconds = 92 miliseconds

Task 3

Determine the outer and inner loop counts in the ARM assembly code below to produce a 0.5-second delay (F = 4 MHz).

```
AREA myCode, CODE, READONLY
EXPORT __main
ENTRY

__main
        PUSH {R4, R5, LR}
        MOV R4, #?
loop1
        MOV R5, #?
loop2
        SUBS R5, R5, #1
        BNE loop2
        SUBS R4, R4, #1
        BNE loop1

        POP {R4, R5, PC}
END
```

Target delay = 0.5 seconds = 0.5 seconds / 1*10^-9 = 500x10^6 nanosecs

Total cycles needed = 500x10^6 nanosecs / 250 nanosecs/cycle = 2x10^6 cycles

Total delay = Outer * Inner * 4 * 250 ns

500,000,000 ns / 1000 ns = Outer * Innner

Outer * Inner = 500,000 ns

Let Outter = 1000

Inner = 500